Sahaj Bhadja

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Summary

Software engineer with 2+ years developing and maintaining **clean, scalable applications** using Java, Python, and JavaScript, with experience across requirements, code review, testing, debugging, deployment, and support.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*

- Developed maintainable Python and FastAPI services across the software development lifecycle, processing 5,000+ monthly jobs with automated validation, error handling, CI/CD deployment, monitoring, and 99%+ production reliability.
- Troubleshot application and database performance issues for 500+ users, reducing latency from 1.2 seconds to 350 milliseconds through systematic debugging, query optimization, indexing, and scalable software improvements.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*

- Built and maintained a full-stack application with Python, Flask, React, and JavaScript, serving 150+ instructors and 10,000+ daily requests while delivering reliable user-facing features in production at scale.
- Implemented concurrent processing, secure file ingestion, automated tests, and model fallbacks, resolving performance and reliability issues while reducing feedback turnaround 60% across text and handwritten submissions at scale.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Developed clean, modular Python pipelines processing 100,000+ trajectory logs, improving long-horizon planning 18% through iterative software design, algorithm evaluation, debugging, and reproducible performance testing across repeatable experiments.
- Collaborated on technical requirements and maintained experiment configurations, evaluation records, and performance documentation, enabling traceable code changes and improving closed-loop system stability 15% during simulation at scale.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Contributed from requirement analysis through deployment for cross-platform microservices, collaborating with Microsoft and GitHub stakeholders while automating 20+ tests across Windows, Linux, and macOS environments in production.
- Reviewed, troubleshot, and optimized Java and Python API components using object-oriented design patterns, improving response times 30%, reducing technical debt 20%, and increasing overall pipeline efficiency 68%.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Developed scalable JavaScript applications using React, Node.js, REST APIs, and eight modular services, synchronizing two million records with 33% fewer conflicts through maintainable software design at scale.
- Supported application deployment across staged environments using Git, Docker, Jenkins, automated tests, and approval gates, cutting deployment time 50% while achieving 95%+ successful build validation across releases.

Skills

Languages: Java, Python, JavaScript, TypeScript, C++, SQL, Shell, Bash
Applications: React, Next.js, Node.js, Express.js, FastAPI, Flask, Spring Boot, REST APIs
Engineering: SDLC, requirements analysis, software design, OOP, code review, debugging, troubleshooting
Quality: Unit testing, integration testing, PyTest, Mockito, Jest, Selenium, build validation
Delivery: Git, GitHub, CI/CD, Jenkins, Docker, AWS, Azure, Linux, multi-environment deployment

Projects

Chat2SQL Code | *Java, JavaScript, Cross-Platform Support*

- Developed a Java and TypeScript database application supporting 10+ platforms, consistently configuring Docker, Maven, Node.js, and Yarn for deployment and debugging across Windows, macOS, and Linux environments.

Linguistix | *Python, Deployment, Validation*

- Developed and deployed a containerized Python REST API on AWS with automated CI/CD releases and validation checks, achieving 87% measured accuracy and production-grade reliability across repeatable environments.

FinNexus Live | *JavaScript, Git, Scalable Applications*

- Built a maintainable Next.js and TypeScript application supporting 50,000+ users and one million transactions with real-time updates, secure APIs, scalable MongoDB storage, Kubernetes, and automated deployment workflows.